

Horizon Project

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Namespace Index

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Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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File Index

4.1 File List

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Chapter 5

Namespace Documentation

5.1 horizon Namespace Reference

Namespaces

- namespace [fd_pattern_graph](#)
- namespace [fds](#)
- namespace [horizon](#)
- namespace [static_fd_analysis](#)
- namespace [utils](#)

5.2 horizon.fd_pattern_graph Namespace Reference

Classes

- class [FDPatternGraph](#)

Variables

- [logger](#) = `get_logger(__name__)`

5.2.1 Detailed Description

`FDPatternGraph`: Build a NetworkX graph from CSV data with Functional Dependency edges.

Each unique (column, value) pair becomes a node. This means:

- Same values in the same column share the same node
- Same values in different columns have different nodes
- Edges represent functional dependencies between column-value pairs

5.2.2 Variable Documentation

5.2.2.1 logger

```
horizon.fd_pattern_graph.logger = get_logger(__name__)
```

5.3 horizon.fds Namespace Reference

Namespaces

- namespace [fd](#)
- namespace [fd_pattern](#)
- namespace [pattern_expression](#)
- namespace [set_of_fds](#)

5.3.1 Detailed Description

Classes relating to functional dependencies.

5.4 horizon.fds.fd Namespace Reference

Classes

- class [FunctionalDependency](#)

5.4.1 Detailed Description

`FunctionalDependency`: Representing a functional dependency with one value on the RHS and potentially multiple

5.5 horizon.fds.fd_pattern Namespace Reference

Classes

- class [FDPattern](#)

5.6 horizon.fds.pattern_expression Namespace Reference

Classes

- class [PatternExpression](#)

5.7 horizon.fds.set_of_fds Namespace Reference

Classes

- class [SetOfFDs](#)

5.8 horizon.horizon Namespace Reference

Functions

- None [repair_tuple](#) (dict[str, list[str]] columns, int t, [FunctionalDependency](#) fd, list[dict[str, str]] repair_table, [PatternExpression](#) p_exp, [FDPatternGraph](#) fd_pattern_graph)
- tuple[list[[PatternExpression](#)], float] [repair_dirty_data](#) (Path dirty_data_path, Path cleaned_data_output_path, list[[FunctionalDependency](#)] ordered_fds, list[dict[str, str]] repair_table, [FDPatternGraph](#) fd_pattern_graph, int|None n_rows=None, bool collect_pattern_expressions=False)
- None [run_horizon](#) (str dataset_name, Path dirty_data_path, [SetOfFDs](#) set_of_fds, Path output_dir, int|None n_rows=None, bool enable_plotting=False, bool collect_pattern_expressions=False)
- None [main](#) (Path dataset_dir, str dirty_data_file, Path output_dir, int|None n_rows, bool enable_plotting, bool collect_pattern_expressions)

Variables

- logging [logger](#) = get_logger(__name__)
- argparse [parser](#)
- [type](#)
- [required](#)
- [help](#)
- [default](#)
- [action](#)
- argparse [args](#) = parser.parse_args()
- [log_level](#)
- [exc_info](#)

5.8.1 Function Documentation

5.8.1.1 main()

```
None horizon.horizon.main (
    Path dataset_dir,
    str dirty_data_file,
    Path output_dir,
    int | None n_rows,
    bool enable_plotting,
    bool collect_pattern_expressions )
```

5.8.1.2 repair_dirty_data()

```
tuple[list[PatternExpression], float] horizon.horizon.repair_dirty_data (
    Path dirty_data_path,
    Path cleaned_data_output_path,
    list[FunctionalDependency] ordered_fds,
    list[dict[str, str]] repair_table,
    FDPatternGraph fd_pattern_graph,
    int | None n_rows = None,
    bool collect_pattern_expressions = False )
```

Repairs data under the given dirty data path, based on the given order and FD Pattern graph.

If n_rows is given, only repairs the first n_rows tuples.

Set collect_pattern_expressions=False to free each tuple's PatternExpression immediately when the lineage is n

Saves cleaned data under given cleaned data output path.

5.8.1.3 repair_tuple()

```
None horizon.horizon.repair_tuple (
    dict[str, list[str]] columns,
    int t,
    FunctionalDependency fd,
    list[dict[str, str]] repair_table,
    PatternExpression p_exp,
    FDPatternGraph fd_pattern_graph )
```

Given a data frame as a dict, applies in-place repairs for tuple with index `t` and functional dependency `fd`.

5.8.1.4 run_horizon()

```
None horizon.horizon.run_horizon (
    str dataset_name,
    Path dirty_data_path,
    SetOfFDs set_of_fds,
    Path output_dir,
    int | None n_rows = None,
    bool enable_plotting = False,
    bool collect_pattern_expressions = False )
```

5.8.2 Variable Documentation

5.8.2.1 action

```
horizon.horizon.action
```

5.8.2.2 args

```
argparse horizon.horizon.args = parser.parse_args()
```

5.8.2.3 default

```
horizon.horizon.default
```

5.8.2.4 exc_info

```
horizon.horizon.exc_info
```

5.8.2.5 help

```
horizon.horizon.help
```

5.8.2.6 log_level

```
horizon.horizon.log_level
```

5.8.2.7 logger

```
logging horizon.horizon.logger = get_logger(__name__)
```

5.8.2.8 parser

```
argparse horizon.horizon.parser
```

Initial value:

```
00001 = argparse.ArgumentParser(  
00002     description="Horizon: Scalable Dependency-driven Data Cleaning."  
00003 )
```

5.8.2.9 required

```
horizon.horizon.required
```

5.8.2.10 type

```
horizon.horizon.type
```

5.9 horizon.static_fd_analysis Namespace Reference

Classes

- class [FDGraph](#)

Functions

- tuple[list[[FunctionalDependency](#)], float] [get_ordered_fds](#) ([SetOfFDs](#) set_of_fds, str dataset_name="", Path output_dir=Path("output"), bool enable_plotting=True)

Variables

- [logger](#) = [get_logger](#)(__name__)

5.9.1 Function Documentation

5.9.1.1 `get_ordered_fds()`

```
tuple[list[FunctionalDependency], float] horizon.static_fd_analysis.get_ordered_fds (
    SetOfFDs set_of_fds,
    str dataset_name = "",
    Path output_dir = Path("output"),
    bool enable_plotting = True )
```

5.9.2 Variable Documentation

5.9.2.1 `logger`

```
horizon.static_fd_analysis.logger = get_logger(__name__)
```

5.10 `horizon.utils` Namespace Reference

Namespaces

- namespace [loaders](#)
- namespace [logging_config](#)

Variables

- list [__all__](#) = ["setup_logging", "get_logger"]

5.10.1 Detailed Description

Horizon utility modules.

5.10.2 Variable Documentation

5.10.2.1 `__all__`

```
list horizon.utils.__all__ = ["setup_logging", "get_logger"] [private]
```

5.11 `horizon.utils.loaders` Namespace Reference

Classes

- class [CSVFDLoader](#)
- class [FDLoader](#)
- class [TXTFDLoader](#)

Functions

- [SetOfFDs get_fds](#) (Path source, list[str]|Path columns_or_path)
- [SetOfFDs load_fds](#) (Path dataset_dir, Path data_csv_path)
- [pl.LazyFrame _scan_csv](#) (Path source, int|None n_rows)
- [pl.LazyFrame _scan_parquet](#) (Path source, int|None n_rows)
- [pl.DataFrame load_table](#) (Path source, list[str]|None columns=None, int|None n_rows=None)
- [pl.LazyFrame lazy_load_table](#) (Path source, list[str]|None columns=None, int|None n_rows=None)
- [Iterator\[pl.DataFrame\] iter_table_batches](#) (Path source, list[str]|None columns=None, int|None n_rows=None, int block_bytes=8 * 1024 * 1024)

Variables

- [logger](#) = get_logger(__name__)
- [dict _SCANNERS](#)

5.11.1 Function Documentation

5.11.1.1 _scan_csv()

```
pl.LazyFrame horizon.utils.loaders._scan_csv (
    Path source,
    int | None n_rows ) [protected]
```

5.11.1.2 _scan_parquet()

```
pl.LazyFrame horizon.utils.loaders._scan_parquet (
    Path source,
    int | None n_rows ) [protected]
```

5.11.1.3 get_fds()

```
SetOfFDs horizon.utils.loaders.get_fds (
    Path source,
    list[str] | Path columns_or_path )
```

Extracts FDs as SetOfFDs from the given source. Source can be fds.csv or fds.txt.

5.11.1.4 iter_table_batches()

```
Iterator[pl.DataFrame] horizon.utils.loaders.iter_table_batches (
    Path source,
    list[str] | None columns = None,
    int | None n_rows = None,
    int block_bytes = 8 * 1024 * 1024 )
```

Yield a table in row batches using memory BOUNDED by block size, not file size.

Additive, non-breaking companion to `lazy_load_table` for streaming a whole table once (e.g. the repair loop). For CSV it reads via `pyarrow`'s block-based reader, whose peak memory is $\sim O(\text{block_bytes})$ regardless of file size -- unlike `scan_csv().collect_batches()`, which under the polars streaming engine materialises ~a full copy of the CSV. Columns are read as Utf8 and lowercased, matching `lazy_load_table`. Pass 'columns' (any casing) to read only a subset -- projection is pushed into the reader so unselected columns are never parsed; missing names raise `ValueError`. Non-CSV sources fall back to the existing lazy streaming path, so their behaviour is unchanged. Stops after `n_rows` rows.

5.11.1.5 lazy_load_table()

```
pl.LazyFrame horizon.utils.loaders.lazy_load_table (
    Path source,
    list[str] | None columns = None,
    int | None n_rows = None )
```

Read a CSV or Parquet table with column names lowercased.
Same as load_table() but returns a LazyFrame.

Allows streaming and is more suitable for data larger than RAM.

5.11.1.6 load_fds()

```
SetOfFDs horizon.utils.loaders.load_fds (
    Path dataset_dir,
    Path data_csv_path )
```

Finds an FD file under the given dataset directory.
Prefers fds.csv, otherwise extracts FDs from fds.txt. Returns SetOfFDs.

5.11.1.7 load_table()

```
pl.DataFrame horizon.utils.loaders.load_table (
    Path source,
    list[str] | None columns = None,
    int | None n_rows = None )
```

Read a CSV or Parquet table with column names lowercased.

Dispatches to a per-extension scanner. Matches the lowercasing in
'CSVFDLoader' so FDs and DataFrame columns share one casing convention.
Pass 'columns' (any casing) to read only a subset -- projection is pushed
down into the scan so unselected columns are never read. Missing names
raise ValueError.

5.11.2 Variable Documentation

5.11.2.1 _SCANNERS

```
dict horizon.utils.loaders._SCANNERS [protected]
```

Initial value:

```
00001 = {
00002     ".csv": _scan_csv,
00003     ".parquet": _scan_parquet,
00004 }
```

5.11.2.2 logger

```
horizon.utils.loaders.logger = get_logger(__name__)
```


5.12 horizon.utils.logging_config Namespace Reference

Functions

- logging.Logger [setup_logging](#) (int log_level=logging.INFO, Optional[Path] log_file=None, Path log_dir=Path("logs"))
- logging.Logger [get_logger](#) (str name)

5.12.1 Detailed Description

Centralized logging configuration for the Horizon pipeline.

Provides consistent logging setup across all modules with:

- File and console logging
- Configurable log levels
- Structured log format with timestamps and module names

5.12.2 Function Documentation

5.12.2.1 [get_logger\(\)](#)

```
logging.Logger horizon.utils.logging_config.get_logger (
    str name )
```

Get a logger for a specific module.

Args:

name: Module name (typically __name__)

Returns:

Configured logger instance

5.12.2.2 [setup_logging\(\)](#)

```
logging.Logger horizon.utils.logging_config.setup_logging (
    int log_level = logging.INFO,
    Optional[Path] log_file = None,
    Path log_dir = Path("logs") )
```

Configure logging for the entire Horizon pipeline.

Args:

log_level: Logging level (logging.DEBUG, logging.INFO, etc.)
log_file: Path to log file. If None, defaults to logs/horizon.log
log_dir: Directory to store log files

Returns:

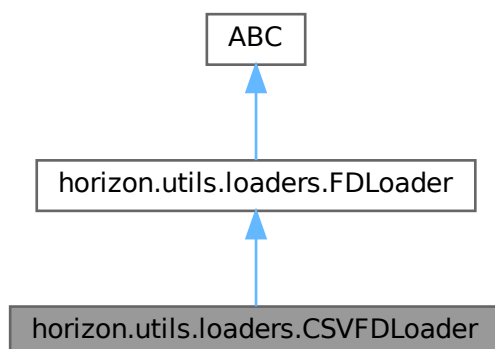
Configured logger instance

Chapter 6

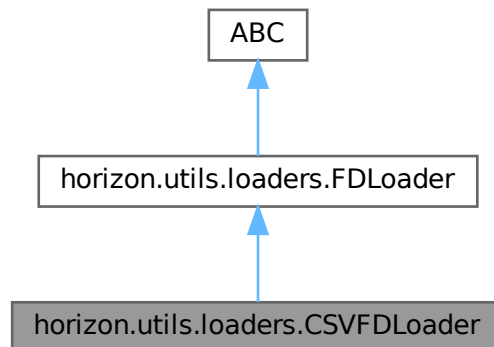
Class Documentation

6.1 horizon.utils.loaders.CSVFDLoader Class Reference

Inheritance diagram for horizon.utils.loaders.CSVFDLoader:



Collaboration diagram for horizon.utils.loaders.CSVFDLoader:



Public Member Functions

- None `__init__` (self)
- [SetOfFDs load](#) (self, Path source)

6.1.1 Detailed Description

Class to load FDs from a CSV file into a `SetOfFDs` object.

6.1.2 Constructor & Destructor Documentation

6.1.2.1 `__init__()`

```
None horizon.utils.loaders.CSVFDLoader.__init__ (  
    self )
```

6.1.3 Member Function Documentation

6.1.3.1 `load()`

```
SetOfFDs horizon.utils.loaders.CSVFDLoader.load (  
    self,  
    Path source )
```

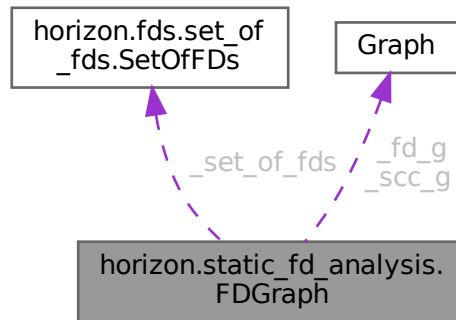
Reimplemented from [horizon.utils.loaders.FDLoader](#).

The documentation for this class was generated from the following file:

- horizon/utils/[loaders.py](#)

6.2 horizon.static_fd_analysis.FDGraph Class Reference

Collaboration diagram for horizon.static_fd_analysis.FDGraph:



Public Member Functions

- None `__init__` (self, [SetOfFDs](#) set_of_fds)
- list[[FunctionalDependency](#)] `order_fds` (self)
- list[int] `dfs_tree` (self, str start_attribute)
- None `plot_graphs` (self, str dataset_name, Path output_dir)
- dict `graph_data` (self)

Protected Member Functions

- Graph `_build_fd_graph` (self)
- Graph `_find_strongly_connected_components` (self)
- list[[FunctionalDependency](#)] `_order_subg` (self, Graph sub_g, int start_v, int order_counter)
- list[[FunctionalDependency](#)] `_get_topological_sorting` (self, Graph g, int order_counter)

Protected Attributes

- `_set_of_fds`
- `_fd_g`
- `_scc_g`
- `_components`

Static Protected Attributes

- [SetOfFDs](#) `_set_of_fds`
- Graph `_fd_g`
- Graph `_scc_g`
- `ig_components`.VertexClustering

6.2.1 Constructor & Destructor Documentation

6.2.1.1 `__init__()`

```
None horizon.static_fd_analysis.FDGraph.__init__ (
    self,
    SetOfFDs set_of_fds )
```

6.2.2 Member Function Documentation

6.2.2.1 `_build_fd_graph()`

```
Graph horizon.static_fd_analysis.FDGraph._build_fd_graph (
    self ) [protected]
```

6.2.2.2 `_find_strongly_connected_components()`

```
Graph horizon.static_fd_analysis.FDGraph._find_strongly_connected_components (
    self ) [protected]
```

Finds strongly connected components and builds SCC graph.

6.2.2.3 `_get_topological_sorting()`

```
list[FunctionalDependency] horizon.static_fd_analysis.FDGraph._get_topological_sorting (
    self,
    Graph g,
    int order_counter ) [protected]
```

Implements Kahn's Algorithm for computing a topological sorting using BFS.

6.2.2.4 `_order_subg()`

```
list[FunctionalDependency] horizon.static_fd_analysis.FDGraph._order_subg (
    self,
    Graph sub_g,
    int start_v,
    int order_counter ) [protected]
```

Implements Hierholzer's linear time algorithm for constructing an Eulerian cycle/tour.

6.2.2.5 dfs_tree()

```
list[int] horizon.static_fd_analysis.FDGraph.dfs_tree (
    self,
    str start_attribute )
```

Constructs a DFS tree and returns all visited edges.

6.2.2.6 graph_data()

```
dict horizon.static_fd_analysis.FDGraph.graph_data (
    self )
```

Structured, JSON-serialisable description of the graphs for the UI.

Takes computed FD graph, SCCG, and FD order data and stores it as JSON data.

6.2.2.7 order_fds()

```
list[FunctionalDependency] horizon.static_fd_analysis.FDGraph.order_fds (
    self )
```

Orders each independent sub-component of the FD graph.

6.2.2.8 plot_graphs()

```
None horizon.static_fd_analysis.FDGraph.plot_graphs (
    self,
    str dataset_name,
    Path output_dir )
```

6.2.3 Member Data Documentation

6.2.3.1 _components [1/2]

```
ig horizon.static_fd_analysis.FDGraph._components .VertexClustering [static], [protected]
```

6.2.3.2 _components [2/2]

```
horizon.static_fd_analysis.FDGraph._components [protected]
```

6.2.3.3 _fd_g [1/2]

```
Graph horizon.static_fd_analysis.FDGraph._fd_g [static], [protected]
```

6.2.3.4 `_fd_g` [2/2]

`horizon.static_fd_analysis.FDGraph._fd_g` [protected]

6.2.3.5 `_scc_g` [1/2]

`Graph horizon.static_fd_analysis.FDGraph._scc_g` [static], [protected]

6.2.3.6 `_scc_g` [2/2]

`horizon.static_fd_analysis.FDGraph._scc_g` [protected]

6.2.3.7 `_set_of_fds` [1/2]

`SetOfFDs horizon.static_fd_analysis.FDGraph._set_of_fds` [static], [protected]

6.2.3.8 `_set_of_fds` [2/2]

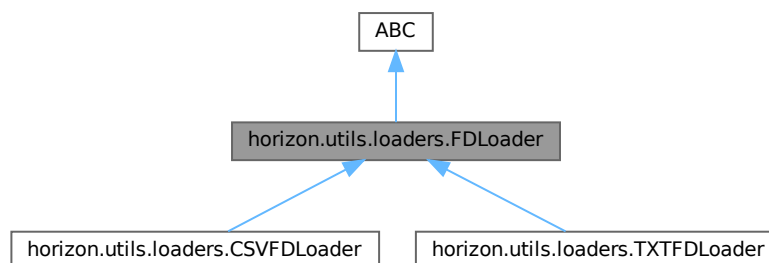
`horizon.static_fd_analysis.FDGraph._set_of_fds` [protected]

The documentation for this class was generated from the following file:

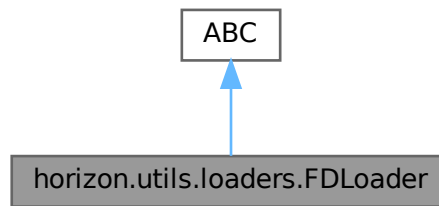
- [horizon/static_fd_analysis.py](#)

6.3 `horizon.utils.loaders.FDLoader` Class Reference

Inheritance diagram for `horizon.utils.loaders.FDLoader`:



Collaboration diagram for horizon.utils.loaders.FDLoader:



Public Member Functions

- [SetOfFDs load](#) (self, source)

6.3.1 Detailed Description

Abstract base class for FD loaders.

6.3.2 Member Function Documentation

6.3.2.1 load()

```

SetOfFDs horizon.utils.loaders.FDLoader.load (
    self,
    source )
  
```

Reimplemented in [horizon.utils.loaders.CSVFDLoader](#), and [horizon.utils.loaders.TXTFDLoader](#).

The documentation for this class was generated from the following file:

- horizon/utils/[loaders.py](#)

6.4 horizon.fds.fd_pattern.FDPattern Class Reference

Public Member Functions

- None [__init__](#) (self, [FunctionalDependency fd](#), tuple|list|str [lval](#), str [rval](#))
- [FunctionalDependency fd](#) (self)
- tuple [lval](#) (self)
- str [rval](#) (self)
- str [__repr__](#) (self)
- bool [__eq__](#) (self, object other)
- int [__hash__](#) (self)

Protected Attributes

- `_fd`
- `_lval`
- `_rval`

6.4.1 Constructor & Destructor Documentation

6.4.1.1 `__init__()`

```
None horizon.fds.fd_pattern.FDPattern.__init__ (  
    self,  
    FunctionalDependency fd,  
    tuple | list | str lval,  
    str rval )
```

6.4.2 Member Function Documentation

6.4.2.1 `__eq__()`

```
bool horizon.fds.fd_pattern.FDPattern.__eq__ (  
    self,  
    object other )
```

6.4.2.2 `__hash__()`

```
int horizon.fds.fd_pattern.FDPattern.__hash__ (  
    self )
```

6.4.2.3 `__repr__()`

```
str horizon.fds.fd_pattern.FDPattern.__repr__ (  
    self )
```

6.4.2.4 `fd()`

```
FunctionalDependency horizon.fds.fd_pattern.FDPattern.fd (  
    self )
```

6.4.2.5 `lval()`

```
tuple horizon.fds.fd_pattern.FDPattern.lval (  
    self )
```

6.4.2.6 rval()

```
str horizon.fds.fd_pattern.FDPattern.rval (
    self )
```

6.4.3 Member Data Documentation

6.4.3.1 _fd

```
horizon.fds.fd_pattern.FDPattern._fd [protected]
```

6.4.3.2 _lval

```
horizon.fds.fd_pattern.FDPattern._lval [protected]
```

6.4.3.3 _rval

```
horizon.fds.fd_pattern.FDPattern._rval [protected]
```

The documentation for this class was generated from the following file:

- [horizon/fds/fd_pattern.py](#)

6.5 horizon.fd_pattern_graph.FDPatternGraph Class Reference

Public Member Functions

- None [__init__](#) (self, Path data_path, [SetOfFDs](#) set_of_fds, str dataset_name="", bool pruning=True, Path output_dir=Path("output"), bool enable_plotting=False)
- list[dict[str, str]] [repair_table](#) (self)
- int [number_of_nodes](#) (self)
- int [number_of_edges](#) (self)
- float [get_edge_quality](#) (self, str from_node, str to_node)
- dict [get_node_info](#) (self, str node_id)
- bool [has_node](#) (self, str col_name, str value)
- tuple[str, str] [choose_best_next_edge](#) (self, str lhs, str lval, str rhs)
- None [clear](#) (self)

Public Attributes

- [graph](#)

Static Public Attributes

- nx [graph](#) .DiGraph

Protected Member Functions

- `str _cell_node_id` (self, str col_name, value)
- `tuple[str, str] _parse_node_id` (self, str node_id)
- `nx.DiGraph _build_graph` (self, Path data_path, [SetOfFDs](#) set_of_fds)
- `None _prune_edges` (self)

Protected Attributes

- `_pruning`
- `_repair_table`

Static Protected Attributes

- `list _repair_table` [dict[str, str]]

6.5.1 Detailed Description

A graph representation of CSV data with functional dependency edges.

Nodes represent unique (column, value) pairs. Multiple cells with the same value in the same column will reference the same node.

6.5.2 Constructor & Destructor Documentation

6.5.2.1 __init__()

```
None horizon.fd_pattern_graph.FDPatternGraph.__init__ (
    self,
    Path data_path,
    SetOfFDs set_of_fds,
    str dataset_name = "",
    bool pruning = True,
    Path output_dir = Path("output"),
    bool enable_plotting = False )
```

Initialize the FDPatternGraph.

Args:

data_path: Path to the CSV file containing the data
 set_of_fds: Parsed set of functional dependencies

6.5.3 Member Function Documentation

6.5.3.1 _build_graph()

```
nx.DiGraph horizon.fd_pattern_graph.FDPatternGraph._build_graph (
    self,
    Path data_path,
    SetOfFDs set_of_fds ) [protected]
```

Build the graph with nodes for unique (column, value) pairs and edges based on functional dependencies. Calculate quality for each edge in the graph and store as edge attributes.

Returns:

NetworkX DiGraph with column-value nodes and FD-based edges

6.5.3.2 _cell_node_id()

```
str horizon.fd_pattern_graph.FDPatternGraph._cell_node_id (
    self,
    str col_name,
    value ) [protected]
```

Generate a unique node ID for a (column, value) pair.

Format: "{col_name}__{value}" - e.g., "brewery-name__Guinness"
 Same value in the same column always produces the same node ID.
 Different columns get different nodes even for the same value.

6.5.3.3 _parse_node_id()

```
tuple[str, str] horizon.fd_pattern_graph.FDPatternGraph._parse_node_id (
    self,
    str node_id ) [protected]
```

Parse a node ID back to (col_name, value).

Args:

node_id: Node identifier like "brewery-name__Guinness"

Returns:

Tuple of (column_name, value)

6.5.3.4 _prune_edges()

```
None horizon.fd_pattern_graph.FDPatternGraph._prune_edges (
    self ) [protected]
```

6.5.3.5 choose_best_next_edge()

```
tuple[str, str] horizon.fd_pattern_graph.FDPatternGraph.choose_best_next_edge (
    self,
    str lhs,
    str lval,
    str rhs )
```

Choose best next edge given a LHS attribute and value, returns RHS attribute and value with the highest quality

6.5.3.6 clear()

```
None horizon.fd_pattern_graph.FDPatternGraph.clear (
    self )
```

Clears the FD Pattern graph.

6.5.3.7 get_edge_quality()

```
float horizon.fd_pattern_graph.FDPatternGraph.get_edge_quality (
    self,
    str from_node,
    str to_node )
```

Get the quality of a specific edge.

Args:

from_node: Source node ID
to_node: Target node ID

Returns:

Quality score for the edge

6.5.3.8 get_node_info()

```
dict horizon.fd_pattern_graph.FDPatternGraph.get_node_info (
    self,
    str node_id )
```

Get all stored information about a node.

6.5.3.9 has_node()

```
bool horizon.fd_pattern_graph.FDPatternGraph.has_node (
    self,
    str col_name,
    str value )
```

True if a (column, value) node exists in the graph.

6.5.3.10 number_of_edges()

```
int horizon.fd_pattern_graph.FDPatternGraph.number_of_edges (
    self )
```

6.5.3.11 number_of_nodes()

```
int horizon.fd_pattern_graph.FDPatternGraph.number_of_nodes (
    self )
```

6.5.3.12 repair_table()

```
list[dict[str, str]] horizon.fd_pattern_graph.FDPatternGraph.repair_table (
    self )
```

6.5.4 Member Data Documentation

6.5.4.1 `_pruning`

`horizon.fds.fd.FunctionalDependency._pruning` [protected]

6.5.4.2 `_repair_table` [1/2]

`list horizon.fds.fd.FunctionalDependency._repair_table` [dict[str, str]] [static], [protected]

6.5.4.3 `_repair_table` [2/2]

`horizon.fds.fd.FunctionalDependency._repair_table` [protected]

6.5.4.4 `graph` [1/2]

`nx horizon.fds.fd.FunctionalDependency.graph` .DiGraph [static]

6.5.4.5 `graph` [2/2]

`horizon.fds.fd.FunctionalDependency.graph`

The documentation for this class was generated from the following file:

- [horizon/fds/fd.py](#)

6.6 horizon.fds.fd.FunctionalDependency Class Reference

Public Member Functions

- `None __init__` (self, tuple|list|str lhs, str rhs, int index=0, int|None order=None)
- `str|tuple lhs` (self)
- `tuple lhs_attributes` (self)
- `str rhs` (self)
- `int index` (self)
- `None index` (self, int index)
- `int|None order` (self)
- `None order` (self, int order)
- `bool|None cyclic` (self)
- `None cyclic` (self, bool cyclic)
- `str __repr__` (self)
- `bool __eq__` (self, object other)
- `int __hash__` (self)
- `tuple as_tuple` (self)
- `list[str] get_attributes` (self)

Public Attributes

- [lhs](#)

Protected Attributes

- [_lhs](#)
- [_rhs](#)
- [_index](#)
- [_order](#)
- [_cyclic](#)

Static Protected Attributes

- `int` [_index](#)
- `int` [_order](#) | `None`
- `bool` [_cyclic](#) | `None`

6.6.1 Constructor & Destructor Documentation

6.6.1.1 `__init__()`

```
None horizon.fds.fd.FunctionalDependency.__init__ (
    self,
    tuple | list | str lhs,
    str rhs,
    int index = 0,
    int | None order = None )
```

6.6.2 Member Function Documentation

6.6.2.1 `__eq__()`

```
bool horizon.fds.fd.FunctionalDependency.__eq__ (
    self,
    object other )
```

6.6.2.2 `__hash__()`

```
int horizon.fds.fd.FunctionalDependency.__hash__ (
    self )
```

6.6.2.3 `__repr__()`

```
str horizon.fds.fd.FunctionalDependency.__repr__ (
    self )
```


6.6.2.4 as_tuple()

```
tuple horizon.fds.fd.FunctionalDependency.as_tuple (
    self )
```

6.6.2.5 cyclic() [1/2]

```
bool | None horizon.fds.fd.FunctionalDependency.cyclic (
    self )
```

6.6.2.6 cyclic() [2/2]

```
None horizon.fds.fd.FunctionalDependency.cyclic (
    self,
    bool cyclic )
```

6.6.2.7 get_attributes()

```
list[str] horizon.fds.fd.FunctionalDependency.get_attributes (
    self )
```

6.6.2.8 index() [1/2]

```
int horizon.fds.fd.FunctionalDependency.index (
    self )
```

6.6.2.9 index() [2/2]

```
None horizon.fds.fd.FunctionalDependency.index (
    self,
    int index )
```

6.6.2.10 lhs()

```
str | tuple horizon.fds.fd.FunctionalDependency.lhs (
    self )
```

6.6.2.11 lhs_attributes()

```
tuple horizon.fds.fd.FunctionalDependency.lhs_attributes (
    self )
```

6.6.2.12 order() [1/2]

```
int | None horizon.fds.fd.FunctionalDependency.order (  
    self )
```

6.6.2.13 order() [2/2]

```
None horizon.fds.fd.FunctionalDependency.order (  
    self,  
    int order )
```

6.6.2.14 rhs()

```
str horizon.fds.fd.FunctionalDependency.rhs (  
    self )
```

6.6.3 Member Data Documentation

6.6.3.1 _cyclic [1/2]

```
bool horizon.fds.fd.FunctionalDependency._cyclic | None [static], [protected]
```

6.6.3.2 _cyclic [2/2]

```
horizon.fds.fd.FunctionalDependency._cyclic [protected]
```

6.6.3.3 _index [1/2]

```
int horizon.fds.fd.FunctionalDependency._index [static], [protected]
```

6.6.3.4 _index [2/2]

```
horizon.fds.fd.FunctionalDependency._index [protected]
```

6.6.3.5 _lhs

```
horizon.fds.fd.FunctionalDependency._lhs [protected]
```

6.6.3.6 _order [1/2]

```
int horizon.fds.fd.FunctionalDependency._order | None [static], [protected]
```

6.6.3.7 `_order` [2/2]

```
horizon.fds.fd.FunctionalDependency._order [protected]
```

6.6.3.8 `_rhs`

```
horizon.fds.fd.FunctionalDependency._rhs [protected]
```

6.6.3.9 `lhs`

```
horizon.fds.fd.FunctionalDependency.lhs
```

The documentation for this class was generated from the following file:

- [horizon/fds/fd.py](#)

6.7 horizon.fds.pattern_expression.PatternExpression Class Reference**Public Member Functions**

- None `__init__` (self, int [tuple_index](#))
- int [tuple_index](#) (self)
- str `__repr__` (self)
- bool `__eq__` (self, object other)
- int `__hash__` (self)
- [FDPattern](#) `__getitem__` (self, int index)
- Iterator[[FDPattern](#)] `__iter__` (self)
- int `__len__` (self)
- None `add_fd_pattern` (self, [FDPattern](#) fd_pattern)
- [FDPattern](#)|None `attribute_in_expression` (self, str attribute)

Protected Attributes

- [_fd_patterns](#)

Static Protected Attributes

- list [_fd_patterns](#) [[FDPattern](#)]

6.7.1 Constructor & Destructor Documentation**6.7.1.1 `__init__()`**

```
None horizon.fds.pattern_expression.PatternExpression.__init__ (
    self,
    int tuple_index )
```

6.7.2 Member Function Documentation

6.7.2.1 `__eq__()`

```
bool horizon.fds.pattern_expression.PatternExpression.__eq__ (
    self,
    object other )
```

6.7.2.2 `__getitem__()`

```
FDPattern horizon.fds.pattern_expression.PatternExpression.__getitem__ (
    self,
    int index )
```

6.7.2.3 `__hash__()`

```
int horizon.fds.pattern_expression.PatternExpression.__hash__ (
    self )
```

6.7.2.4 `__iter__()`

```
Iterator[FDPattern] horizon.fds.pattern_expression.PatternExpression.__iter__ (
    self )
```

6.7.2.5 `__len__()`

```
int horizon.fds.pattern_expression.PatternExpression.__len__ (
    self )
```

6.7.2.6 `__repr__()`

```
str horizon.fds.pattern_expression.PatternExpression.__repr__ (
    self )
```

6.7.2.7 `add_fd_pattern()`

```
None horizon.fds.pattern_expression.PatternExpression.add_fd_pattern (
    self,
    FDPattern fd_pattern )
```

6.7.2.8 `attribute_in_expression()`

```
FDPattern | None horizon.fds.pattern_expression.PatternExpression.attribute_in_expression (
    self,
    str attribute )
```

6.7.2.9 tuple_index()

```
int horizon.fds.pattern_expression.PatternExpression.tuple_index (
    self )
```

6.7.3 Member Data Documentation

6.7.3.1 _fd_patterns [1/2]

```
list horizon.fds.pattern_expression.PatternExpression._fd_patterns [FDPattern] [static],
[protected]
```

6.7.3.2 _fd_patterns [2/2]

```
horizon.fds.pattern_expression.PatternExpression._fd_patterns [protected]
```

The documentation for this class was generated from the following file:

- [horizon/fds/pattern_expression.py](#)

6.8 horizon.fds.set_of_fds.SetOfFDs Class Reference

Public Member Functions

- None [__init__](#) (self, list[[FunctionalDependency](#)]|None [set_of_fds](#), set[str]|None [bound_attributes](#)=None)
- list[[FunctionalDependency](#)] [set_of_fds](#) (self)
- set[str] [unique_attributes](#) (self)
- set[str] [bound_attributes](#) (self)
- None [bound_attributes](#) (self, set[str] [bound_attributes](#))
- set[str] [source_attributes](#) (self)
- None [source_attributes](#) (self, set[str] [source_attributes](#))
- list[[FunctionalDependency](#)] [get_ordered_set_of_fds](#) (self)
- None [add_fd](#) (self, [FunctionalDependency](#) fd)
- list[tuple] [as_tuple_list](#) (self)
- bool [is_bound](#) (self, str attribute)
- bool [is_source](#) (self, str attribute)
- str [__repr__](#) (self)
- bool [__eq__](#) (self, object other)
- int [__hash__](#) (self)
- [FunctionalDependency](#) [__getitem__](#) (self, int index)
- Iterator[[FunctionalDependency](#)] [__iter__](#) (self)
- int [__len__](#) (self)

Protected Attributes

- [_set_of_fds](#)
- [_bound_attributes](#)
- [_source_attributes](#)
- [_unique_attributes](#)

Static Protected Attributes

- list [_set_of_fds](#) [[FunctionalDependency](#)]
- set [_unique_attributes](#) [str]
- set [_bound_attributes](#) [str]
- set [_source_attributes](#) [str]

6.8.1 Constructor & Destructor Documentation

6.8.1.1 `__init__()`

```
None horizon.fds.set_of_fds.SetOfFDs.__init__ (
    self,
    list[FunctionalDependency] | None set_of_fds,
    set[str] | None bound_attributes = None )
```

6.8.2 Member Function Documentation

6.8.2.1 `__eq__()`

```
bool horizon.fds.set_of_fds.SetOfFDs.__eq__ (
    self,
    object other )
```

6.8.2.2 `__getitem__()`

```
FunctionalDependency horizon.fds.set_of_fds.SetOfFDs.__getitem__ (
    self,
    int index )
```

6.8.2.3 `__hash__()`

```
int horizon.fds.set_of_fds.SetOfFDs.__hash__ (
    self )
```

6.8.2.4 `__iter__()`

```
Iterator[FunctionalDependency] horizon.fds.set_of_fds.SetOfFDs.__iter__ (
    self )
```

6.8.2.5 `__len__()`

```
int horizon.fds.set_of_fds.SetOfFDs.__len__ (
    self )
```

6.8.2.6 `__repr__()`

```
str horizon.fds.set_of_fds.SetOfFDs.__repr__ (
    self )
```

6.8.2.7 `add_fd()`

```
None horizon.fds.set_of_fds.SetOfFDs.add_fd (
    self,
    FunctionalDependency fd )
```

Adds a new FD, adjusting its index (to preserve uniqueness), and updating unique attributes.

6.8.2.8 `as_tuple_list()`

```
list[tuple] horizon.fds.set_of_fds.SetOfFDs.as_tuple_list (
    self )
```

Returns the set of FDs as tuples

6.8.2.9 `bound_attributes()` [1/2]

```
set[str] horizon.fds.set_of_fds.SetOfFDs.bound_attributes (
    self )
```

6.8.2.10 `bound_attributes()` [2/2]

```
None horizon.fds.set_of_fds.SetOfFDs.bound_attributes (
    self,
    set[str] bound_attributes )
```

6.8.2.11 `get_ordered_set_of_fds()`

```
list[FunctionalDependency] horizon.fds.set_of_fds.SetOfFDs.get_ordered_set_of_fds (
    self )
```

Returns ordered set of FDs, set by static_fd_analyis.py.

6.8.2.12 `is_bound()`

```
bool horizon.fds.set_of_fds.SetOfFDs.is_bound (
    self,
    str attribute )
```

True if attribute is bound.

6.8.2.13 is_source()

```
bool horizon.fds.set_of_fds.SetOfFDs.is_source (
    self,
    str attribute )
```

True if attribute never appears as an FD RHS (a source column).

6.8.2.14 set_of_fds()

```
list[FunctionalDependency] horizon.fds.set_of_fds.SetOfFDs.set_of_fds (
    self )
```

6.8.2.15 source_attributes() [1/2]

```
set[str] horizon.fds.set_of_fds.SetOfFDs.source_attributes (
    self )
```

6.8.2.16 source_attributes() [2/2]

```
None horizon.fds.set_of_fds.SetOfFDs.source_attributes (
    self,
    set[str] source_attributes )
```

6.8.2.17 unique_attributes()

```
set[str] horizon.fds.set_of_fds.SetOfFDs.unique_attributes (
    self )
```

6.8.3 Member Data Documentation**6.8.3.1 _bound_attributes** [1/2]

```
set horizon.fds.set_of_fds.SetOfFDs._bound_attributes [str] [static], [protected]
```

6.8.3.2 _bound_attributes [2/2]

```
horizon.fds.set_of_fds.SetOfFDs._bound_attributes [protected]
```

6.8.3.3 _set_of_fds [1/2]

```
list horizon.fds.set_of_fds.SetOfFDs._set_of_fds [FunctionalDependency] [static], [protected]
```


6.8.3.4 `_set_of_fds` [2/2]

`horizon.fds.set_of_fds.SetOfFDs._set_of_fds` [protected]

6.8.3.5 `_source_attributes` [1/2]

`set horizon.fds.set_of_fds.SetOfFDs._source_attributes` [str] [static], [protected]

6.8.3.6 `_source_attributes` [2/2]

`horizon.fds.set_of_fds.SetOfFDs._source_attributes` [protected]

6.8.3.7 `_unique_attributes` [1/2]

`set horizon.fds.set_of_fds.SetOfFDs._unique_attributes` [str] [static], [protected]

6.8.3.8 `_unique_attributes` [2/2]

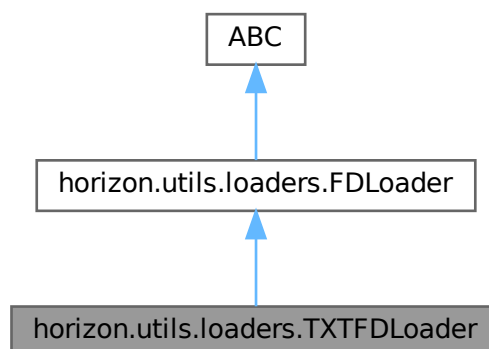
`horizon.fds.set_of_fds.SetOfFDs._unique_attributes` [protected]

The documentation for this class was generated from the following file:

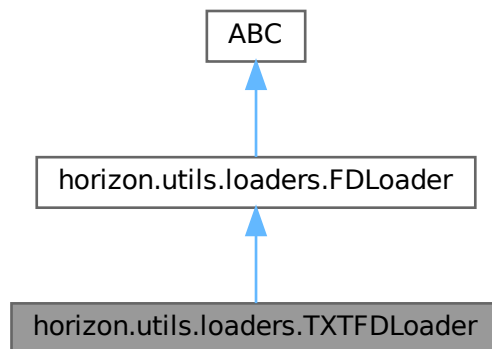
- `horizon/fds/set_of_fds.py`

6.9 horizon.utils.loaders.TXTFDLoader Class Reference

Inheritance diagram for `horizon.utils.loaders.TXTFDLoader`:



Collaboration diagram for horizon.utils.loaders.TXTFDLoader:



Public Member Functions

- None `__init__` (self, list[str]|Path columns_or_path, str separator="->")
- list[str] `load_column_names` (self, Path columns_csv_path)
- tuple[list[str], str]|None `parse_line` (self, line)
- SetOfFDs `load` (self, Path source)

Protected Attributes

- `_column_names`

Static Protected Attributes

- dict `_EXTENSION_LOADERS`

6.9.1 Detailed Description

Class to load FDs from a TXT file into a SetOfFDs object. Assumes FDs in the file are represented by column indices and a separator (e.g., 1 -> 5).

6.9.2 Constructor & Destructor Documentation

6.9.2.1 `__init__()`

```

None horizon.utils.loaders.TXTFDLoader.__init__ (
    self,
    list[str] | Path columns_or_path,
    str separator = "->" )

```

Constructor requires a list of columns or a CSV path from which the column names can be extracted. It also requires a separator (default: ->).

6.9.3 Member Function Documentation

6.9.3.1 load()

```
SetOfFDs horizon.utils.loaders.TXTFDLoader.load (  
    self,  
    Path source )
```

Reimplemented from [horizon.utils.loaders.FDLoader](#).

6.9.3.2 load_column_names()

```
list[str] horizon.utils.loaders.TXTFDLoader.load_column_names (  
    self,  
    Path columns_csv_path )
```

Loads column names from a given CSV file and returns them as a list.

6.9.3.3 parse_line()

```
tuple[list[str], str] | None horizon.utils.loaders.TXTFDLoader.parse_line (  
    self,  
    line )
```

Helper function to parse one line of a file and translate it into a tuple containing the FD's LHS and RHS.

6.9.4 Member Data Documentation

6.9.4.1 _column_names

```
horizon.utils.loaders.TXTFDLoader._column_names [protected]
```

6.9.4.2 _EXTENSION_LOADERS

```
dict horizon.utils.loaders.TXTFDLoader._EXTENSION_LOADERS [static], [protected]
```

Initial value:

```
= {  
    ".csv": CSVFDLoader,  
    ".txt": TXTFDLoader,  
}
```

The documentation for this class was generated from the following file:

- [horizon/utils/loaders.py](#)

Chapter 7

File Documentation

7.1 horizon/fd_pattern_graph.py File Reference

Classes

- class [horizon.fd_pattern_graph.FDPatternGraph](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.fd_pattern_graph](#)

Variables

- [horizon.fd_pattern_graph.logger](#) = `get_logger(__name__)`

7.2 horizon/__init__.py File Reference

7.3 horizon/fds/__init__.py File Reference

Namespaces

- namespace [horizon](#)
- namespace [horizon.fds](#)

7.4 horizon/utils/__init__.py File Reference

Namespaces

- namespace [horizon](#)
- namespace [horizon.utils](#)

Variables

- list [horizon.utils.__all__](#) = ["setup_logging", "get_logger"]

7.5 horizon/fds/fd.py File Reference

Classes

- class [horizon.fds.fd.FunctionalDependency](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.fds](#)
- namespace [horizon.fds.fd](#)

7.6 horizon/fds/fd_pattern.py File Reference

Classes

- class [horizon.fds.fd_pattern.FDPattern](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.fds](#)
- namespace [horizon.fds.fd_pattern](#)

7.7 horizon/fds/pattern_expression.py File Reference

Classes

- class [horizon.fds.pattern_expression.PatternExpression](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.fds](#)
- namespace [horizon.fds.pattern_expression](#)

7.8 horizon/fds/set_of_fds.py File Reference

Classes

- class [horizon.fds.set_of_fds.SetOfFDs](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.fds](#)
- namespace [horizon.fds.set_of_fds](#)

7.9 horizon/horizon.py File Reference

Namespaces

- namespace [horizon](#)
- namespace [horizon.horizon](#)

Functions

- None [horizon.horizon.repair_tuple](#) (dict[str, list[str]] columns, int t, [FunctionalDependency](#) fd, list[dict[str, str]] repair_table, [PatternExpression](#) p_exp, [FDPatternGraph](#) fd_pattern_graph)
- tuple[list[[PatternExpression](#)], float] [horizon.horizon.repair_dirty_data](#) (Path dirty_data_path, Path cleaned_data_output_path, list[[FunctionalDependency](#)] ordered_fds, list[dict[str, str]] repair_table, [FDPatternGraph](#) fd_pattern_graph, int|None n_rows=None, bool collect_pattern_expressions=False)
- None [horizon.horizon.run_horizon](#) (str dataset_name, Path dirty_data_path, [SetOfFDs](#) set_of_fds, Path output_dir, int|None n_rows=None, bool enable_plotting=False, bool collect_pattern_expressions=False)
- None [horizon.horizon.main](#) (Path dataset_dir, str dirty_data_file, Path output_dir, int|None n_rows, bool enable_plotting, bool collect_pattern_expressions)

Variables

- logging [horizon.horizon.logger](#) = get_logger(__name__)
- argparse [horizon.horizon.parser](#)
- [horizon.horizon.type](#)
- [horizon.horizon.required](#)
- [horizon.horizon.help](#)
- [horizon.horizon.default](#)
- [horizon.horizon.action](#)
- argparse [horizon.horizon.args](#) = parser.parse_args()
- [horizon.horizon.log_level](#)
- [horizon.horizon.exc_info](#)

7.10 horizon/static_fd_analysis.py File Reference

Classes

- class [horizon.static_fd_analysis.FDGraph](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.static_fd_analysis](#)

Functions

- tuple[list[[FunctionalDependency](#)], float] [horizon.static_fd_analysis.get_ordered_fds](#) ([SetOfFDs](#) set_of_fds, str dataset_name="", Path output_dir=Path("output"), bool enable_plotting=True)

Variables

- [horizon.static_fd_analysis.logger](#) = `get_logger(__name__)`

7.11 horizon/utils/loaders.py File Reference

Classes

- class [horizon.utils.loaders.FDLoader](#)
- class [horizon.utils.loaders.CSVFDLoader](#)
- class [horizon.utils.loaders.TXTFDLoader](#)

Namespaces

- namespace [horizon](#)
- namespace [horizon.utils](#)
- namespace [horizon.utils.loaders](#)

Functions

- [SetOfFDs](#) [horizon.utils.loaders.get_fds](#) (Path source, list[str]|Path columns_or_path)
- [SetOfFDs](#) [horizon.utils.loaders.load_fds](#) (Path dataset_dir, Path data_csv_path)
- pl.LazyFrame [horizon.utils.loaders._scan_csv](#) (Path source, int|None n_rows)
- pl.LazyFrame [horizon.utils.loaders._scan_parquet](#) (Path source, int|None n_rows)
- pl.DataFrame [horizon.utils.loaders.load_table](#) (Path source, list[str]|None columns=None, int|None n_rows=None)
- pl.LazyFrame [horizon.utils.loaders.lazy_load_table](#) (Path source, list[str]|None columns=None, int|None n_rows=None)
- Iterator[pl.DataFrame] [horizon.utils.loaders.iter_table_batches](#) (Path source, list[str]|None columns=None, int|None n_rows=None, int block_bytes=8 * 1024 * 1024)

Variables

- [horizon.utils.loaders.logger](#) = `get_logger(__name__)`
- dict [horizon.utils.loaders._SCANNERS](#)

7.12 horizon/utils/logging_config.py File Reference

Namespaces

- namespace [horizon](#)
- namespace [horizon.utils](#)
- namespace [horizon.utils.logging_config](#)

Functions

- logging.Logger [horizon.utils.logging_config.setup_logging](#) (int log_level=logging.INFO, Optional[Path] log_file=None, Path log_dir=Path("logs"))
- logging.Logger [horizon.utils.logging_config.get_logger](#) (str name)

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